SDUHSD CAREER TECHNICAL EDUCATION

AP Computer Science Principles

Campus: Canyon Crest Academy
CTE Industry Sector: Information and Communication Technologies
CTE Pathway: Software and Systems Development
Course Sequence: Course 2 - Concentrator
Prerequisites: Recommended Computer Science Foundations or Computer Game Design
10 credits, practical art or elective, weighted grade, UC Approved Area G
Recommended Pathway Sequence:

Course 1	Course 2	Course 3	Enrichment
Computer Science Foundations OR Computer Game Design	Human-Computer Interaction (HCI) Design		AP Computer Science Principles AP Computer Science A Machine Learning

Course Description:
AP Computer Science Principles offers a multidisciplinary approach to teaching the underlying principles of computation. The course will introduce students to the creative aspects of programming, abstractions, algorithms, large data sets, the Internet, cybersecurity concerns, and computing impacts. AP Computer Science Principles also gives students the opportunity to use current technologies to create computational artifacts for both self-expression and problem solving. Together, these aspects of the course make up a rigorous and rich curriculum that aims to broaden participation in computer science. AT CCA we explore these topics while working with App Inventor and Android devices.

Course Syllabi:

CCA-APCSP

Grading:
For the through-course assessment, you will upload digital artifacts and written responses via a Web-based digital application. You will be asked to describe or analyze your work, whether it includes research, the creation of an artifact (e.g., a video, spreadsheet, graph, or electronic slide show),

or the creation of a program.. The end-of-course AP Exam will be a paper and pencil written exam. It will be 120 minutes long and will include approximately 74 multiple-choice questions. There are two types of multiple-choice questions:

- Single Select Multiple-Choice: you select 1 answer from among 4 options
- Multiple Select Multiple-Choice: you select 2 answers from among 4 options

Work load: (High, Medium, Low)
Class setting: (This relates directly to how much the student is expected to work independently or in a group requiring self-initiated work, problem solving, etc versus and highly guided and structured class). (Highly independent, Moderately Independent, Structured)

*Note: Each year all CTE courses are reviewed by industry professionals within the pathway to ensure the most current, relevant experience for our students.

Textbooks or Supplemental Materials for this course:

CCA - Remington - Mobile CSP - Online Textbook

Software, program knowledge specific to this course:

CCA - Remington - MIT App Inventor 2

Industry Connections:

Sony Interactive Entertainment - Development Operations Manager - Sue Xu

Units of Study, Large Projects, Work-based Learning Connections.

Included in Syllabus - CCA-Remington-APCSP

The following Model Curriculum Standards connected to this course:

Standards for Career Ready Practice
1. Apply appropriate technical skills and academic knowledge. 4. Apply technology to enhance productivity. 5. Utilize critical thinking to make sense of problems and persevere in solving them.

10. Demonstrate creativity and innovation.

Framework Concept	Software and Systems Development Standards
C1.0 Identify and apply the systems development process.	C1.1 Identify the phases of the systems development life cycle, including analysis, design, programming, testing, implementation, maintenance, and improvement. C1.4 Work as a member of, and within the scope and boundaries of, a development project team.
C3.0 Create effective interfaces between humans and technology.	C3.1 Describe and apply the basic process of input, processing, and output. C3.2 Design effective and intuitive interfaces using knowledge of cognitive, physical, and social interactions.
C4.0 Develop software using programming languages.	C4.1 Identify and describe the abstraction level of programming languages from low-level, hardware-based languages to high-level, interpreted, Web-based languages. C4.3 Identify and use different authoring tools and integrated development environments (IDEs). C4.7 Use various data structures, arrays, objects, files, and databases. C4.9 Create programs using control structures, procedures, functions, parameters, variables, error recovery, and recursion. C4.10 Create and know the comparative advantages of various queue, sorting, and searching algorithms. C4.11 Document development work for various audiences, such as comments for other programmers, and manuals for users.
C5.0 Test, debug, and improve software development work.	C5.4 Test software and projects. C5.5 Evaluate results against initial requirements.
C6.0 Integrate a variety of media into development projects.	C6.1 Identify the basic design elements necessary to produce effective print, video, audio, and interactive media. C6.4 Develop a presentation or other multimedia project: video, game, or interactive Web sites, from storyboard to production. C6.6 Integrate media into a full project using appropriate tools. C6.7 Create and/or capture professional-quality media, images, documents, audio, and video clips.
C8.0 Develop databases.	C8.1 Describe the critical function of databases in modern organizations.